NOTE

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1 Motivation

application: fault tolerant quantum computing.

connection between phase of matter and the application on quantum computer ?

how to find the error from measurement ?

An TO-trivial phase transition occurs at a critical value of the error rate; if the error rate is below the critical value (the accuracy threshold), encoded information can be protected arbitrarily well in the limit of a large code block. find this threshold ?

A measure error (boundary error) comes from multiple possible bulk errors: $P(m) = \sum_E P(m|E)P(E)$

$E_{virtual} -> E_{physical} -> E(m)$

2 Method

2.1 isoTNS representation

We consider the toric code on a square lattice with open boundary conditions,

$$H = -\sum_s A_s - \sum_p B_p, \qquad A_s = \prod_{i \in s} \sigma_i^z, \qquad B_p = \prod_{i \in p} \sigma_i^x.$$

Each bond is represented by a doubled Hilbert space, $|i\rangle \mapsto |ii\rangle$. The local rank-eight tensor is

$$T^{ijkl}_{\alpha\beta\gamma\delta} = \begin{cases} \frac{1}{\sqrt{2}}\,\delta_{i\alpha}\delta_{j\beta}\delta_{k\gamma}\delta_{l\delta}, & \text{if } \alpha+\beta+\gamma+\delta = 0 \pmod 2, \\ 0, & \text{otherwise.} \end{cases}$$

The physical and virtual indices are ordered as east, north, west, and south.

Independent bit-flip errors occur on the internal virtual bonds with probability p. A distance-d patch contains $2d(d-1)$ such bonds.

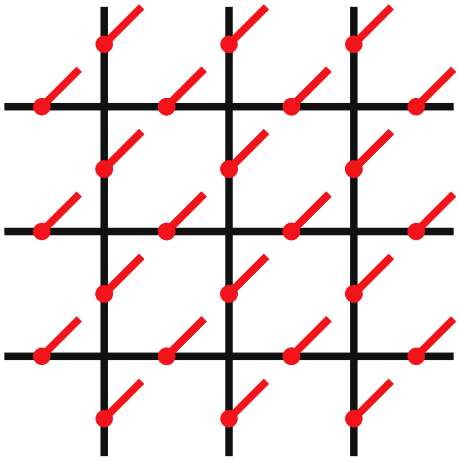


Figure 1: Square-lattice geometry of the planar code with open boundaries.

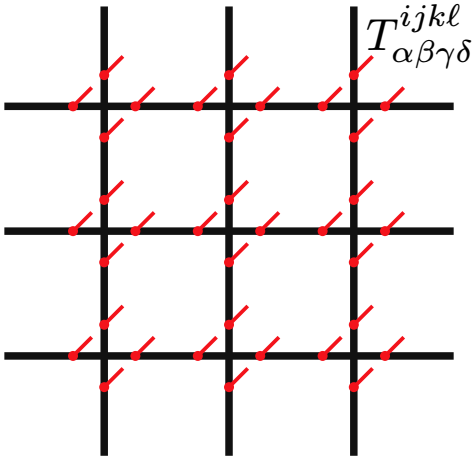


Figure 2: Planar-code lattice in the doubled-Hilbert-space representation. Each virtual bond is doubled as $|i\rangle \mapsto |ii\rangle$, and $T^{ijkl}_{\alpha\beta\gamma\delta}$ labels the local rank-eight tensor.

2.2 Isometric tensors and sequential circuit

2.2.1 Local isometries

The local tensor can be regarded as an isometry and completed to a unitary gate. This correspondence provides a circuit representation of the tensor network.

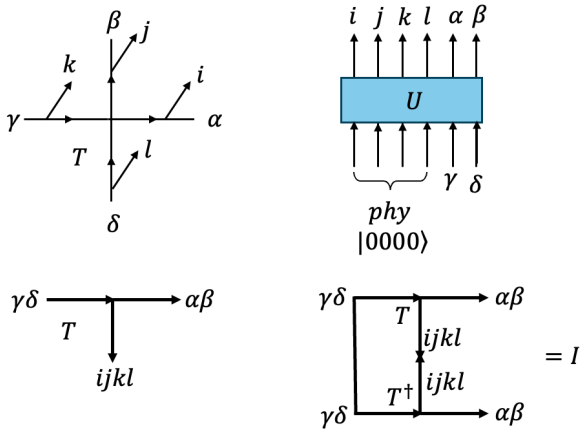


Figure 3: Local tensor T as an isometry and its completion to a unitary gate U. The input legs γ, δ, together with four physical ancillas initialized in $|0000\rangle$, are mapped to the output legs $\alpha, \beta, i, j, k, l$; the lower panels show the reshaped tensor and the isometry condition $T^\dagger T = I$.

2.2.2 Boundary conditions and gate schedule

The planar patch has two rough and two smooth boundaries and encodes one logical qubit. Physical qubits are initialized in $|0\rangle$. The virtual boundary convention is $|+\rangle$ on the west and east boundaries and $|0\rangle$ on the south and north boundaries. At the east and north boundaries, the terminal carrier is absorbed into the local gate.

There are four local-gate types: a six-qubit bulk gate, five-qubit east and north boundary gates, and a four-qubit corner gate. Each gate satisfies the isometry condition. Gates are applied in diagonal layers, progressing from south to north and from west to east.

2.3 Sampling and virtual-bond diagnostics

The circuit admits a classical message-passing description. The Yao.jl implementation is a small-system reference: it applies each local gate to a state vector, performs a Born measurement, records the result, and resets the measured physical qubits. The production estimator instead works directly with batches of virtual-bond error frames. For each shot, it samples independent horizontal and vertical bond-error bits with probability p, then evaluates the detector record and west-boundary logical parity by XOR operations. The full detector batch is decoded once with PyMatching, and a

logical failure is counted when the predicted logical parity differs from the sampled one. No unitary gates or individual physical measurement outcomes are simulated.

2.3.1 Virtual-bond noise checks

For an $L \times L$ lattice, with $L = d$, the following quantities diagnose the virtual-bond noise and its percolation behaviour:

1. *Injected-error and mismatch densities:*
$$\rho_{\text{injected}} = \frac{\text{number of erroneous virtual bonds}}{2L(L-1)+4L} \approx p,$$
$$\rho_{\text{mismatch}} = \frac{\text{number of mismatched internal doubled bonds}}{2L(L-1)} \approx p.$$

2. *Plaquette frustration:* the probability that a plaquette has odd error parity is
$$P_{\text{odd}} = \binom{4}{1} p(1-p)^3 + \binom{4}{3} p^3(1-p) = \frac{1}{2}\left[1-(1-2p)^4\right].$$

3. *Largest occupied cluster:* the normalized cluster size is $S_{\max}/L^2$, where $S_{\max}$ is the number of sites in the largest connected error cluster.

4. *Horizontal spanning probability:*
$$P_{\text{span}}^{\text{horizontal}} = P(\text{an error cluster connects the west and east boundaries}).$$

2.4 Syndrome extraction

A detector is a binary indicator of whether an error violates an expected parity constraint. Let $\mathbf{e}$ denote the binary virtual-bond error vector. The detector record is

$$\mathbf{s} = \mathsf{H}\mathbf{e} \pmod 2,$$

where H is the parity-check matrix. Equivalently,

$$s_i = \begin{cases} 0, & \text{if an even number of adjacent bonds have errors,} \\ 1, & \text{if an odd number of adjacent bonds have errors.} \end{cases}$$

For bit-flip noise, we use two detector families:

1. an elementary plaquette detector,
$$S_{r,c} = h_{r,c} \oplus h_{r+1,c} \oplus v_{r,c} \oplus v_{r,c+1};$$

2. north and south boundary detectors,
$$n_c = h_{1,c}, \qquad s_c = h_{d,c}.$$

The total number of detectors is $(d-1)^2 + 2(d-1) = d^2 - 1$.

2.5 Decoding and logical-failure estimation

PyMatching returns both a minimum-weight correction and the logical parity associated with that correction. Let E be the physical error, C the decoder correction, and $R = E \oplus C$ the residual error. A logical failure occurs when

$$l(R) = l(E) \oplus l(C) = 1.$$

The finite-size scaling ansatz for the logical-failure probability is

$$P_{\text{fail}}(d, p) \simeq F\Big((p - p_c)d^{1/\nu}\Big).$$

For each (d, p) point and each batch, we:

1. sample all $2d(d-1)$ virtual-bond errors for multiple shots;
2. compute the $d^2 - 1$ detector values algebraically;
3. compute the hidden west-boundary logical parity;
4. decode the batch with PyMatching; and
5. count disagreements between the actual and predicted logical parities.

For N shots with F logical failures,

$$P_{\text{fail}} = \frac{F}{N}, \qquad \text{SE}(P_{\text{fail}}) = \sqrt{\frac{P_{\text{fail}}(1 - P_{\text{fail}})}{N}}.$$

2.6 Bootstrap uncertainty

To quantify the uncertainty of a crossing estimate, we partition the shots at each (d, p) point into equal batches. Each bootstrap replicate resamples these batch failure rates with replacement, recomputes the monotone logical-failure curves for adjacent code distances, and extracts their unique crossing. The confidence interval is obtained from the corresponding empirical quantiles of the valid bootstrap crossings.

3 compile with Clifford gate

To encode the $|0_L\rangle$ state, we propose two ways:

1. based on $|0_L\rangle \propto (I + \bar{Z}) \prod_s (I + A_s)|+\rangle^{\otimes N}$

2. based on $|0_L\rangle \propto \prod_p (I + B_p)|0\rangle^{\otimes N}$

Here is an $3 * 3$ example:

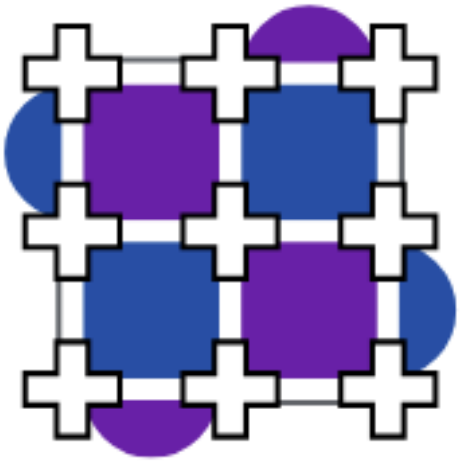


Figure 4: $3 * 3$ lattice. purple: A_s, blue: B_p

1. A_s-based encoding:

(a) A CNOT can establish an even-parity constraint. eg:

$$Z_2 Z_3 Z_5 Z_6 = +1 \quad \Longleftrightarrow \quad x_2 \oplus x_3 \oplus x_5 \oplus x_6 = 0.$$

If qubit 5 starts in $|0\rangle$, the CNOTs

$$2 \to 5, \qquad 3 \to 5, \qquad 6 \to 5$$

compute $x_5 = x_2 \oplus x_3 \oplus x_6$, establishing that parity.

(b) choose qubit 2 to fix the logical-zero parity, 4,5,7,9 as fresh parity targets. These qubits start in $|0\rangle$;

leaves $1, 3, 6, 8$ as independent inputs, these qubits start in $|+\rangle$, so we apply

$$H_1, \ H_3, \ H_6, \ H_8.$$

(c) $\bar{Z} = Z_1 Z_2 Z_3$. Apply

$$1 \to 2, \qquad 3 \to 2.$$

Since x_2 started at zero, it becomes $a \oplus b$. Therefore,

$$x_1 \oplus x_2 \oplus x_3 = a \oplus (a \oplus b) \oplus b = 0.$$

This establishes $\bar{Z} = +1$.

(d) Build the four As constraints block by block.

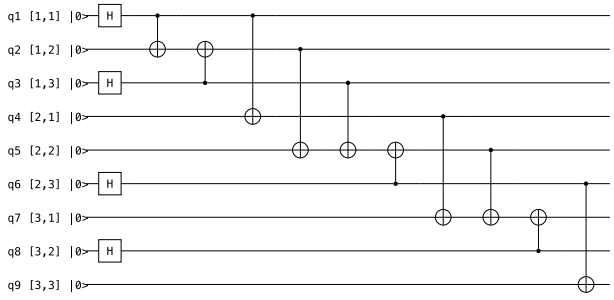


Figure 5: A_s-based encoding for $3 * 3$ lattice.

2. B_p-based encoding:

choose represensitive qubits for every B_p plaquette, and apply H, others are target qubits.

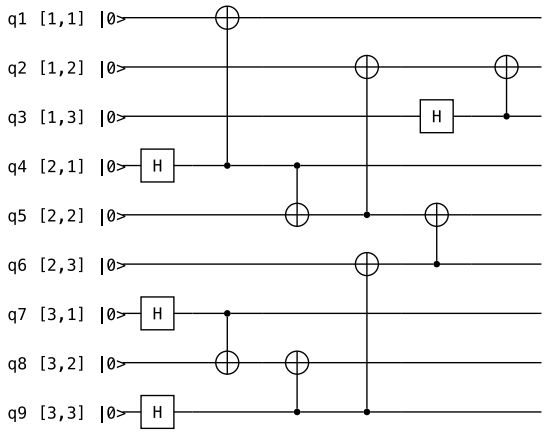


Figure 6: B_p-based encoding for $3 * 3$ lattice.

4 error model

apply X-only or Z-only error after each elementary gate (H, CNOT) with rate p

5 sampling

Monte Carlo sampling chooses which faults occur; the Pauli-frame simulator determines their effect on the circuit.

For each code distance d and physical error probability p:

1. Sample faults. After every gate, independently apply an X fault with probability p to each participating qubit. Each CNOT therefore has two possible fault locations.

2. Propagate errors. Track the X/Z error pattern through the remaining circuit using exact Clifford rules,

$$E \longrightarrow UEU^{\dagger}.$$

This accounts for errors spreading and cancelling, without storing the full wavefunction.

3. Decode and judge success. Calculate the final, ideal stabilizer outcomes and give them to the matching decoder. Compare its proposed correction with the actual propagated error. For preparation of $|0_L\rangle$, a remaining logical X counts as failure.

4. Repeat and estimate. Run N independent trials and count F failures:

$$\boxed{\widehat{P}_L(d,p) = \frac{F}{N}.}$$

Error propagation is exact for the Clifford circuit and Pauli-noise model; the estimated failure probability has finite-sampling uncertainty.

6 results(2D)

1. The two Z-only panels

The Z-only noise at most cause Logical-Z faults

$$\bar{Z}|0_L\rangle = |0_L\rangle.$$

Logical-Z faults can occur without changing the $|0_L\rangle$ state. Maybe need the logical-Z failure curves?

2. The X-only panels

The As/X is fragile logical-parity preparation. Single X fault during logical-state preparation can become a complete logical X error eg, in Figure 4, only X_2 would cause a whole logical-X string.

In the parity part, for m independent locations, their odd-flip probability is

$$q_m(p) = \frac{1-(1-2p)^m}{2},$$

which imposes a logical-failure floor. Increasing m makes this floor larger, approaching 1/2, not smaller.

contain more CNOTs, so more fragile to X-only noise.

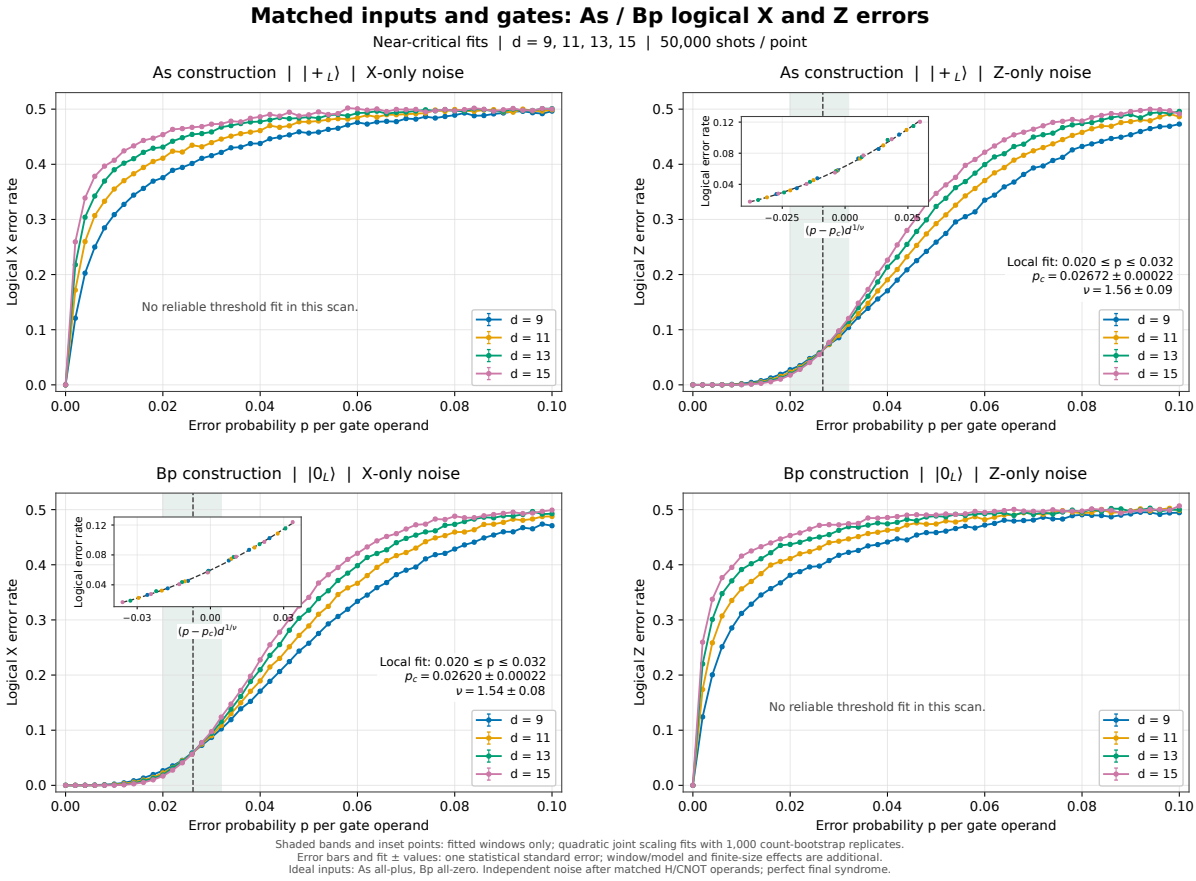


Figure 7: X error and Z error. As based and Bp based circuit is just for different basis, they are symmetry. Bp/X: X propagate only form short-range error; Bp/Z: Z propagate form long-range error.

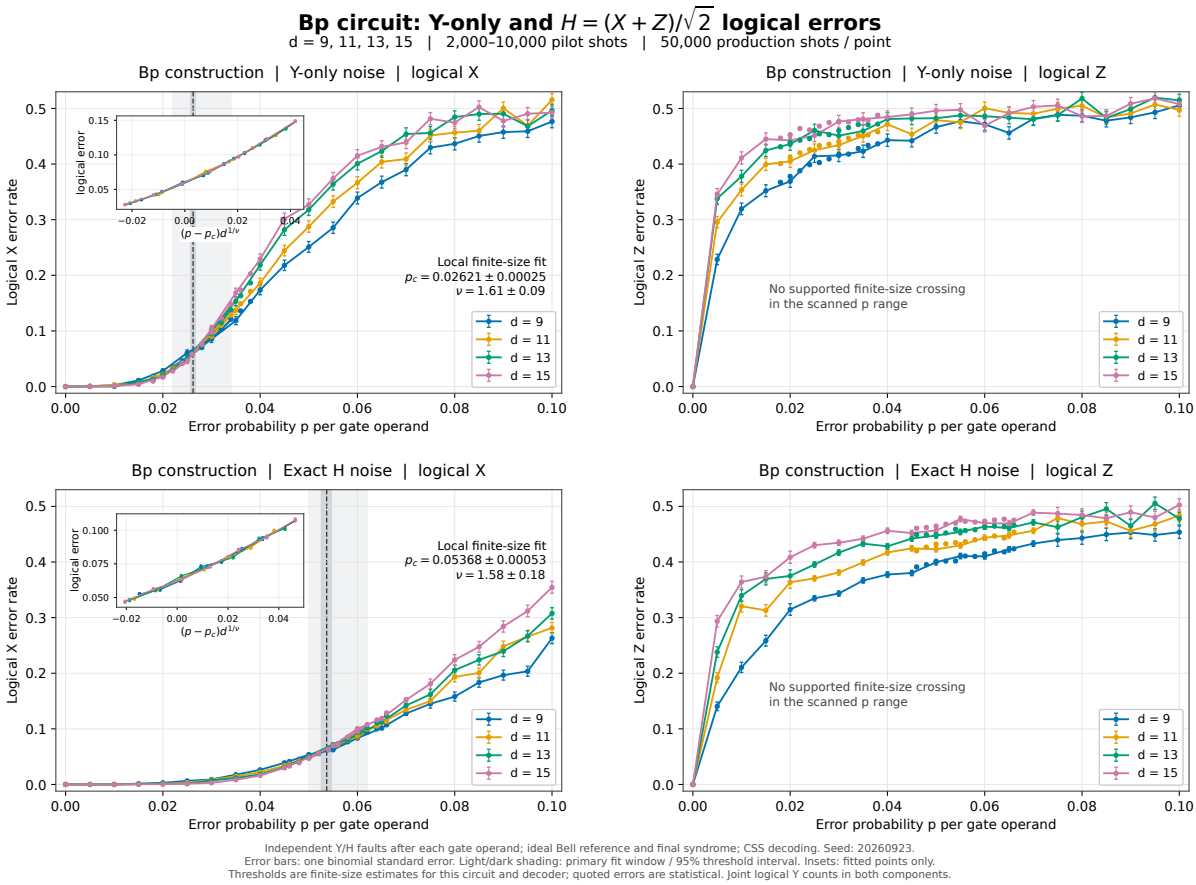


Figure 8: Y error and (X+Z)/$\sqrt{(2)}$ error. Y: seperate decoder treat X and Z separately. (X+Z)/$\sqrt{(2)}$: rotate an angle only change prefactor, the logical Z threshold would always be 0 only if Z element is not zero.

7 3D toric code

7.1 Geometry, checks, and logical operators

We use a relative cubic slab with periodic y, z directions and rough open x ends. There are L_x retained vertex planes $x = 1, \ldots, L_x$; x-edges lie at $x = 0, \ldots, L_x$, while transverse edges lie at $x = 1, \ldots, L_x$. Coordinates $y \in \mathbb{Z}_{L_y}$ and $z \in \mathbb{Z}_{L_z}$ are periodic. The two periodic length-two cells are retained as distinct cells, and boundary plaquettes are triangular. We allow $L_x \geq 1$ and $L_y, L_z \geq 2$. With $M = L_y L_z$,

$$H = -\sum_v A_v - \sum_f B_f, \qquad A_v = \prod_{e \ni v} X_e, \quad B_f = \prod_{e \in \partial f} Z_e, \qquad n = (3L_x + 1)M.$$

There are $N_A = L_x M$ independent stars and $N_B = (3L_x + 2)M$ plaquettes of rank $(2L_x + 1)M - 1$. Thus $k = n - N_A - \operatorname{rank} B = 1$. For example, the two rough-end xy checks are

$$B^{xy}_{0,y,z} = Z^x_{0,y,z} Z^x_{0,y+1,z} Z^y_{1,y,z}, \qquad B^{xy}_{L_x,y,z} = Z^x_{L_x,y,z} Z^x_{L_x,y+1,z} Z^y_{L_x,y,z}.$$

The xz end checks follow by exchanging y and z; interior plaquettes have four edges. The $2 \times 2 \times 2$ reference therefore has 28 qubits, 8 independent stars, and 32 plaquettes of rank 19. Its circuit has 8 Hadamards and 40 CNOTs, with a checkpoint after each four-star plane. A longitudinal Z string joining the rough ends has distance $d_Z = L_x + 1$; a transverse X membrane has distance $d_X = M$. The prepared state is the normalized projector state

$$|0_L\rangle = 2^{N_A/2} \prod_v \frac{I + A_v}{2} |0\rangle^{\otimes n}.$$

7.2 Sequential circuit, noise, and feedback

Visit vertices in increasing (x, y, z) order. At each vertex the fresh outgoing x-edge is Hadamard transformed, then controls CNOTs onto the incoming x-edge and the $+y, -y, +z, -z$ edges, in that order. The last outgoing x-edges terminate at the right rough end. For $L_y = 2$ or $L_z = 2$, opposite directions are still distinct edge qubits. We assume ideal initialization, idle periods, stabilizer measurement, and correction. Each active output of an H or CNOT independently acquires a Bernoulli-p X error in the X-only experiment, or a Bernoulli-p Z error in the Z-only experiment. The inactive Pauli channel has rate zero.

Both protocols replay the same sampled gate faults. Final-only recovery decodes the full syndrome once after preparation. Layer feedback measures and decodes after every completed x layer, correcting only qubits in the current prefix slab. Its frontier has moving triangular plaquettes; future qubits are unavailable for correction. At each checkpoint, a Z recovery from star defects uses matching and an X recovery from plaquette defects uses belief propagation with ordered-statistics decoding. Matching is only the Z-recovery component, not a decoder for both CSS channels. The small-size exact option minimizes data-recovery weight; it is not exact circuit-level maximum-likelihood inference. Decoder ties, finite BP iterations, and the ideal measurement assumption limit conclusions from the curves.

The reported residual logical-X and logical-Z parities are Pauli-frame diagnostics. Either-logical failure is their union; state failure for the prepared $|0_L\rangle$ is logical-X failure alone. In particular a logical Z changes a Pauli diagnostic but leaves $|0_L\rangle$ unchanged.

7.3 An indistinguishable Z-fault pair

Assume $0 \le p < 1/2$ and a decoder with access only to measured stabilizer syndromes and declared model parameters. For each transverse position (y, z) choose two independent Z-fault bits: a on the left boundary x-edge after its first CNOT, and b on the outgoing x-edge after the final CNOT of the first-plane gadget. At checkpoint j, a alone leaves Z on the $x = 0$ edge; b alone has propagated to the $x = 1, \ldots, j$ edges. These two errors give the same first-plane star syndrome. Toggling both adds $D_j = Z_{x=0} Z_{x=1} \cdots Z_{x=j}$, a full prefix longitudinal string invisible to every checkpoint syndrome. At the final checkpoint D_{L_x} is logical Z. The $M = L_y L_z$ pairs are disjoint. Give a decoder a genie that reveals every other fault and each pair XOR $a \oplus b$, the syndrome information, but not the constituent bits. The remaining ambiguity in a toggles the final logical-Z parity. If any XOR is one, its two possible bit assignments have equal probability, so the parity is unbiased. All XORs vanish with probability t^M, where $t = (1-p)^2 + p^2$. Conditioned on a zero XOR, the two assignments $(a, b) = (0, 0)$ and $(1, 1)$ have probabilities $(1-p)^2/t$ and p^2/t. Their parity bias is $(1-2p)/t$; across all M pairs it is $((1-2p)/t)^M$. Even this genie-aided decoder therefore has error probability

$$P_{Z,\text{diagnostic}} \ge \frac{1 - t^M [(1-2p)/t]^M}{2} = \frac{1 - (1-2p)^M}{2}.$$

This is a lower bound for the logical-Z diagnostic under the specified gate faults and schedule. It is not a fidelity bound for $|0_L\rangle$: partial Z strings can stabilize the prepared state, and logical Z itself acts trivially on that state.

7.4 Numerical comparison

The default scan uses cubic $L = 3, 5, 7$, ten physical rates from zero to 0.15, and 1000 paired fault records per point. It reports both distances, logical-X, logical-Z, either-logical and state failure, Wilson intervals, and paired protocol differences with uncertainty. A timed pilot on the largest size at $p = 0.03$ precedes the full run. The figure overlays the analytic Z bound; it is a diagnostic reference, not a threshold fit.

A second scan resolves the original Cartesian x-sweep using 10000 paired fault records at each point, with

$$p \in \{10^{-6}, 3\times10^{-6}, 10^{-5}, 3\times10^{-5}, 10^{-4}, 3\times10^{-4}\}, \qquad L \in \{3, 5, 7, 9\}.$$

Let $p_{1\%}(L)$ denote the linearly interpolated physical rate at which the logical-Z diagnostic reaches one percent. With layer feedback the measured values are

$$\big(p_{1\%}(3), p_{1\%}(5), p_{1\%}(7), p_{1\%}(9)\big) = (1.39\times10^{-4}, 5.03\times10^{-5}, 2.92\times10^{-5}, 1.51\times10^{-5}).$$

A four-size power-law summary gives $p_{1\%}(L) \propto L^{-1.97}$, with a binomial-bootstrap exponent interval 1.77–2.17. Thus the apparent small finite-size crossing moves toward zero rather than approaching a positive threshold, as required by the decoder-independent bound above. For example, already at $p = 10^{-5}$ the feedback diagnostic rises from 0.06% at $L = 3$ to 0.73% at $L = 9$.

7.5 Direct vertex-projector H+5 experiment

The primary new diagonal experiment applies the literal cubic-star projector. This is a paper-motivated construction, not a reproduction of an explicit three-dimensional gate circuit supplied by the paper. Set $t = x + y + z$, $u = x$, $v = y$, with open $t = 0, \ldots, T$ and periodic $(u, v) \in \mathbb{Z}_{L_u} \times \mathbb{Z}_{L_v}$. An edge $E[d, t, r]$ ends at $V(t, r)$ and begins at $V(t-1, r-s_d)$, where $s_1 = (1, 0)$, $s_2 = (0, 1)$, $s_3 = (0, 0)$. The encoded slab has

$$n = 3(T+1)L_uL_v, \qquad k = 2, \qquad d_Z = 2\min(L_u, L_v), \quad d_X = (T+1)\min(L_u, L_v).$$

Distinct physical edges remain distinct when a transverse period is two. For the old vertex $V(t-1, r)$, let $e_d = E[d, t-1, r]$ and $n_d = E[d, t, r+s_d]$. Apply, in this order,

$$H(n_1), \quad \mathrm{CX}(n_1, n_2), \quad \mathrm{CX}(n_1, n_3), \quad \mathrm{CX}(n_1, e_1), \quad \mathrm{CX}(n_1, e_2), \quad \mathrm{CX}(n_1, e_3).$$

There are six gates and eleven participating outputs per cell. The independent post-output Bernoulli-p X-only or Z-only noise convention is unchanged; initialization, idles, boundary operations, measurement, and correction are ideal.

Product zero fixes both logical-Z cycles. Ideally project the independent lower white X triads and correct their outcomes. After the bulk circuit, project the upper X triads. Their clean outcomes b are uniform on the even subspace: draw $L_uL_v - 1$ bits and fix the last parity. Let u be the flips induced by the noisy Z frame and let $a = b \oplus u$ be the observed outcome. Only a enters the physical boundary controller. With its deterministic correction map C, the relative clean/noisy frame is

$$z' = z \oplus C(a) \oplus C(b), \qquad a = b \oplus u.$$

The map uses physical-edge-ID-ordered breadth-first paths to bulk root $A_{T-1,0,0}$ after deleting the union of both declared logical-X supports. This exclusion guarantees zero logical overlap of every open correction path; acyclicity alone does not. The graph stays connected through direction-three vertical edges and nonwrapping transverse edges. This is a fixed correction convention and changes noisy logical-Z diagnostics compared with an arbitrary uncut tree. The clean b and gate faults remain hidden from all decoders.

Every immutable gate-fault record and clean boundary branch is reused by four arms. Intermediate feedback occurs only after $t = 1, \ldots, T-1$:

1. `none`: boundary completion and final full-code recovery only.
2. `one_step`: in X, measure the periodic six-edge Q_t checks, pair point defects by unit-length MWPM, and apply the first and last edge of each canonical path (a length-one path is applied once). Old-target X faults are unseen but cannot propagate further. Remaining Q_t defects receive one contraction, not a repeated moving-frontier transport. In Z, match the retiring/frontier strip (W, A_0) at $t = 1$ or (A_{t-2}, A_{t-1}) later. Odd parity receives exactly one future dummy; even parity receives none. Apply an endpoint edge only at a retiring endpoint, so every retiring syndrome clears while frontier defects may wait. Symmetric endpoint trimming is not causal in this strip.
3. `window_complete`: restrict recovery to $E_{t-1} \cup E_t$. X uses BP+OSD on P_1 initially and $P_{t-1} \cup P_t$ subsequently. Z uses ordinary open-boundary MWPM on the retiring/frontier strip. Both clear all completed checks.

4. `full_prefix`: recover on all $E_0, \ldots, E_t$ from the completed plaquettes or lower and bulk stars. Future data are unavailable.

The arms differ in measurements, residual constraints, decoder, and correction support; feedback-versus-none differences do not isolate correction range.

All arms use common final BP+OSD X recovery and MWPM Z recovery. The fixed BP baseline is minimum sum, 50 iterations, scale one, parallel schedule, one thread, and `OSD_CS` order two, with uniform prior $p_{\text{eff}} = \min(1 - 10^{-12}, \max(10^{-12}, p))$. PyMatching 2.4.0 and ldpc 2.4.1 are pinned. Matching first minimizes physical length and then lexicographically fixes endpoint/path signatures, retaining period-two parallel edges. These decoders omit circuit-induced correlations. Every complete recovery must satisfy $Hc = s$; a partial route instead obeys $s' = s \oplus Hr$ with its causal residual constraint. A syndrome-valid recovery need not restore the intended logical state.

Score both transverse logical-X bits, both logical-Z bits, and all sixteen Pauli classes. Either-pair logical X is also $|00_L\rangle$ preparation failure for the X-only experiment. Logical Z is a state-independent Pauli diagnostic, not $|00_L\rangle$ infidelity. The four-arm statistics retain sixteen joint masks per observable, Wilson intervals, and all six paired contrasts. Atomic checkpoints reproduce complete records, including b. A separate timed pilot precedes a 500-shot scout at $L = 3, 5, 7$. The scout freezes channel-specific production grids and fit windows before 1000-shot production. Threshold claims require adjacent-size crossing consistency, an interior monotone likelihood fit, acceptable goodness of fit, and at least 80% usable bootstrap fits. Failed guards leave diagnostic candidates.

Preliminary measurements at three sizes. The user-confirmed run completed $L = 3, 5, 7$ on 14 October 2026: 60 X-only and 135 Z-only points, with 1000 paired records per point, 195000 records and 780000 arm replays in total. There were no invalid recoveries. Recorded sampling, decoder-building and replay times were 1532.72 seconds for X and 12168.14 seconds for Z; these include the original cost of reused records and exclude warmup, compilation, plotting and deferred sizes. They are not the wall time of the last resumed segment.

No H+5 threshold or scaling exponent is estimated. All 14 frozen candidate windows fail the unchanged minimum-four-size guard before fitting. The requested 1000 bootstrap replicates were therefore not executed, and no fitted interval or sensitivity fit is reported. This limitation is not evidence for a zero threshold.

Table 1 gives illustrative points selected after the run from the frozen grid. Each rate is based on 1000 paired records. For X at $L = 7$, $p = 0.039448164$, one-step feedback increases failure by 4.4 percentage points relative to none (per-contrast 95% interval $[1.38, 7.38]$), and either complete protocol increases it by 10.8 points ($[6.99, 14.50]$). At the higher sampled rate $p = 0.077446809$, the window-complete change is instead -11.4 points ($[-17.07, -5.62]$), and the full-prefix change is -11.7 points ($[-17.34, -5.94]$). Thus finite-size feedback effects depend on the error rate.

At the lowest sampled Z rate $p = 0.0096470592$, window-complete changes the diagnostic by -10.6 points at $L = 3$ ($[-15.39, -5.70]$), but by -4.6 points at $L = 7$ ($[-10.10, 0.94]$). Most Z curves are near the 0.75 union probability of two uniform logical bits. The grid thus supplies little unsaturated evidence at larger sizes. Paired intervals use simultaneous bounds on the two discordance proba-

Channel	p	L	None	One step	Window	Prefix
X	0.039448164	3	13.6	14.8	17.7	18.2
X	0.039448164	5	6.5	12.3	17.1	17.1
X	0.039448164	7	4.2	8.6	15.0	15.0
Z	0.0096470592	3	51.6	50.2	41.0	41.1
Z	0.0096470592	5	74.5	75.5	64.7	64.7
Z	0.0096470592	7	76.3	74.4	71.7	71.7

Table 1: Preliminary H+5 logical-union rates in percent at $L = 3, 5, 7$. X denotes logical-zero preparation failure; Z is a state-independent logical-Pauli diagnostic, not logical-zero infidelity.

bilities within each contrast, without multiple-comparison coverage across rates, sizes or contrasts. These illustrative comparisons are exploratory, not a confirmatory threshold test.

The original six-size scan was deliberately stopped at 106350 records, and its four-size precursor at 206000 records, after runtime-directed user amendments. Their immutable selections and checkpoints are preserved; $L = 9, 11, 13$ remain deferred. The final subset reused 186000 records exactly and completed 9000 more. Point seeds retain their parent indices in the 260-element `Random.Xoshiro(1234)` UInt64 stream; no existing record was reseeded. The three-size selection hash begins `1597b9941132d3f2`. All original rates, candidate windows, decoder settings and scientific source hashes were preserved. Full hashes, commands and checkpoint ancestry appear in the repository's `preliminary-l3-7/README.md` and H+5 report.

Independent audits verified all joint counts, paired statistics, logical labels, checkpoint digests, reused states and source/data hashes, with warmup excluded. Fresh regressions passed 45610 Julia assertions across 145 test sets and all 51 Python tests. Measured curves, six paired contrasts per point and the guarded diagnostic report are saved in the repository's `preliminary-l3-7/` result directory; the H+5 report gives full paths. Historical H+6 values below belong to a separate experiment and are not substituted for H+5 measurements.

7.6 Historical plaquette-transfer H+6 Z-sector estimate

The historical seven-gate plaquette-transfer H+6 circuit avoids the earlier Cartesian propagation mechanism by using diagonal slices and transferring from old control qubits to new target triads. Every single Z gate-output fault then has final support on at most three qubits in two adjacent slices. To define a logical observable, use a periodic honeycomb toric-code seed instead of fixing a unique product boundary state. Apply noisy H gates to its $3L_yL_z$ qubits, then ideally measure its Z-type compatibility checks and apply X corrections. These checks commute with the specified independent seed Z faults, so they prepare the intended logical $|++\rangle$ code space without removing those faults; no perfect entangled seed is assumed. The encoded slab has

$$n = 3(T+1)L_yL_z, \qquad k = 2, \qquad d_Z = 2\min(L_y, L_z), \qquad d_X = (T+1)\min(L_y, L_z).$$

H+5/H+6 threshold values are not a controlled circuit-only comparison: the seed preparation, upper boundary instrument, and historical channel-dependent seed noise also differ. The saved H+6 data and interfaces remain unchanged.

The transfer retains independent rate-p faults on every active H/CNOT output and ideal measurements, corrections, and idles. Final unit-weight matching on the star syndrome returns a full

data recovery, whose syndrome is checked before scoring either transverse logical-Z winding. For this Z-only preparation of $|++\rangle$, a residual logical Z changes the state. The optional feedback in this older scan also measures Z-type compatibility checks and applies X corrections. It is therefore inactive under pure Z noise, so its result agrees exactly with final-only recovery in this sector. This inactive operation is distinct from the X-star measurements and Z corrections used by the later three-strength Z experiment.

The completed scan uses $T = L_y = L_z = L = 9, 13, 17, 21, 25$ and 10,000 shots per point, for 330,000 production shots. A quadratic logistic finite-size fit with a fixed leading $1/L$ correction gives

$$p_{c,Z} \simeq 0.0052 \quad \text{(approximately 0.52\% per active gate output)}.$$

Fits with different size cuts, probability windows, and correction choices give estimates roughly 0.49%–0.54%; this is a sensitivity range, not a confidence interval. The statistical bootstrap, Wilson intervals, raw counts, model diagnostics, and reproducible commands are recorded in `toric3d-z-threshold.md` in `docs/superpowers` and the associated results. This is a decoder-dependent Z-sector estimate for the diagonal circuit; it neither removes the earlier rough-slab bound nor establishes a joint X/Z threshold.

7.7 Historical plaquette-transfer H+6 X-sector estimate

To test the measurement-feedback mechanism itself, use the same encoded diagonal geometry but prepare logical $|00\rangle$. Product zero already fixes the seed Z compatibility checks and both logical-Z cycles. Ideally measure the independent seed X stars and apply a syndrome-valid Z correction. Under the present convention this preparation has no noisy seed gate: only the thirteen participating outputs of each seven-gate transfer cell acquire independent X faults of rate p.

Before transfer t, measure the six-edge Z compatibility checks on active slice $t-1$. Their X-error defects are point excitations on a periodic triangular matching graph. Unit-weight MWPM pairs the defects. The primary one-step rule moves each endpoint one graph edge toward its partner and maps this virtual old-slice repair through the fixed transfer to a correction on the fresh targets. Thus many X errors are removed before they can act as controls in later transfers. The classical matching computation is global; this implementation is not a constant-depth spatially local channel.

After the final slice, measure all physical Z plaquettes and apply a validated BP+OSD X recovery using minimum-sum BP, 50 iterations, parallel updates, scaling factor one, one thread, and `OSD_CS` order two. The decoder uses a uniform physical-p data prior and does not see the sampled fault record. State failure is the union of the two residual logical-X membrane parities. Unlike the rough-slab logical-Z diagnostic, this event changes the prepared $|00\rangle$ state.

The production scan uses

$$L \in \{5, 7, 9, 11, 13\}, \qquad p \in \{0.0100, 0.0125, 0.0150, 0.0175, 0.0200, 0.0225, 0.0250\},$$

with $T = L_y = L_z = L$ and 1000 paired fault records per point, for 35000 unique production shots. Final-only failure increases with size throughout this window and has no adjacent-size crossing. One-step feedback reverses the size dependence near $p = 0.015$. For example, at $p = 0.015$ the

feedback failure rates for $L = 7, 9, 11, 13$ are respectively $0.248, 0.233, 0.211, 0.226$, whereas the corresponding final-only rates are $0.287, 0.399, 0.464, 0.591$.

The all-size quadratic finite-size fit is rejected by its deviance test. An all-size cubic alternative gives candidate $p_c = 0.01470$ but narrowly misses the preset goodness guard. The least aggressive accepted sensitivity uses the cubic scaling function and $L \geq 7$:

$$p_{c,X} = 0.01482, \qquad \nu = 1.056, \qquad p_{\text{deviance}} = 0.01996.$$

Of 1000 parametric-bootstrap refits, 999 pass the numerical checks, giving the conditional 95% interval

$$p_{c,X} \in [0.01428, 0.01543].$$

Accepted size and probability-window variants give p_c from 0.01476 to 0.01511. The appropriate conclusion is therefore an X-only threshold of approximately 1.5% per participating gate output, conditional on the accepted finite-size model and the ideal-feedback assumptions. A 200-shot full-path routing check at $L = 13, p = 0.015$ gives failure 0.200, compared with 0.226 for one-step routing, but its uncertainty is too large for a precise routing comparison.

This experiment implements the paper's qualitative diagonal-slice mechanism: measure stabilizers in a two-dimensional slice, pair point defects, and feed repairs forward before continuing the sweep. The encoded boundary, transfer gates, tie breaks, post-gate noise, and final BP+OSD decoder are explicit choices made here, so the number is not a reproduction of a threshold quoted by the paper. It also does not combine this $|00\rangle$ X-sector result with the preceding $|++\rangle$ Z-sector estimate into a joint Pauli threshold. Raw counts, Wilson intervals, rejected fits, bootstrap samples, hashes, and reproduction commands are recorded in `toric3d-x-feedback-threshold.md` in `docs/superpowers` and the associated results directory.

7.8 Historical plaquette-transfer H+6 with three feedback strengths

To hold the diagonal sweep fixed while comparing feedback policies, a further paired experiment uses the same encoded diagonal transfer and compares three protocols. The *no-feedback* protocol performs no intermediate correction and invokes only the common final decoder. For X errors, the *slice-local* protocol measures the old active slice's Z compatibility checks before the next transfer. It computes complete MWPM paths, retains their endpoint steps, and applies the propagation image of this partial route to the fresh targets after the noisy transfer. This is target feed-forward, rather than a direct data recovery on the old slice. Its *full-prefix* counterpart measures all completed prefix Z plaquettes after every nonfinal transfer and applies a complete BP+OSD X recovery anywhere in the prepared prefix.

For Z errors, the local and full-prefix protocols first measure both colors of seed X stars and apply a complete MWPM Z recovery on the seed edges; the no-feedback protocol omits this seed-time feedback. After each nonfinal transfer, the local protocol measures a syndrome-complete moving band of three X-star check sheets and applies a complete MWPM Z recovery on the newest two data slices. A literal one-slice Z window would be invalid for the stated post-gate noise model, because faults on old CNOT controls can place a defect on the next older check sheet. The full-prefix Z protocol instead measures all completed prefix X stars and may apply its complete MWPM

Z recovery anywhere in the prepared prefix. Unlike the inactive Z-type compatibility operation in the older Z-only scan, these X-star measurements detect and correct pure Z faults.

For each channel and Monte Carlo shot, one immutable fault record is replayed through all three protocols. Their eight joint Boolean outcomes therefore give both protocol marginals and paired differences for the full implemented policies. In particular, the Z comparison includes the seed-time feedback difference described above and does not isolate spatial support alone. Any complete intermediate or final decoder recovery that fails to reproduce its measured syndrome aborts the point. The deliberately partial X one-step feed-forward need not clear that syndrome, but its underlying complete MWPM route is validated before truncation. The X study prepares logical $|00\rangle$ with no noisy seed gates and scores either residual logical-X membrane. The Z study prepares logical $|++\rangle$ with an independent Z fault after every seed H and scores either residual logical-Z winding. Thus each reported channel observable is a state failure, but the two channel thresholds also differ in their logical input, seed circuit, fault locations, and decoder.

The local protocol has total check participation $O(TL_yL_z)$, whereas repeatedly decoding the growing prefix costs $O(T^2L_yL_z)$. For X noise, the one-step route can leave residual active-slice syndrome, while full-prefix BP+OSD clears the complete prefix syndrome. For Z noise, the three-sheet local protocol already restores zero complete-prefix star syndrome at every checkpoint. Full-prefix Z recovery therefore changes the available paths and logical-class/tie choices, and need not outperform the local Z protocol. The X comparison changes the check and decoder policy as well as correction support, so a threshold increase cannot be attributed to support range alone.

For the cubic specialization below, $T = L_y = L_z = L$. The frozen target grid requested 1000 paired shots for X at

$$L = 5, 7, 9, 11, 13, \qquad p = 0.025, 0.030, 0.040, 0.045, 0.050, 0.055, 0.060, 0.070, 0.080,$$

and 2000 paired shots for Z at

$$L = 9, 13, 17, 21, 25, \qquad p = 0.025, 0.030, 0.0325, 0.035, 0.0375, 0.040, 0.045, 0.050, 0.060, 0.070.$$

Measured runtime sets the final size scope to four complete sizes per channel. X uses $L = 5, 7, 9, 11$ at all nine X probabilities, totaling 36 size-rate points and 36000 paired fault records. Z uses $L = 9, 13, 17, 21$ at the first seven probabilities through $p = 0.045$, totaling 28 points and 56000 paired fault records. This preserves a guard point above the transition and both endpoint-trimmed window checks; the higher frozen Z rates chiefly probe saturation and were stopped before fitting on the measured runtime. The partial $L = 13$ X and $L = 25$ Z checkpoints are retained only as resumable provenance and are excluded from every fit and aggregate.

A deliberate interruption revealed a now-fixed bookkeeping bug: a user interrupt was temporarily labeled as an invalid recovery, and scalar X replay could record several outcomes before its batch completion counter advanced. The original interruption artifacts are archived separately. The live checkpoints were repaired from atomic counts or deterministic replay, and the partial X $L = 13$, Z $L = 25$, and original-shard Z $L = 21, p = 0.045$ records are not analysis inputs. These interruption records are not decoder failures.

Thresholds are exposed only when every adjacent size pair crosses in the fit window, the monotone likelihood fit passes numerical and deviance checks, and at least 80% of 1000 conditional bootstrap

fits are usable. Conditional Monte Carlo intervals and the spread among accepted size, window, polynomial, and fixed-L^{-1}-correction sensitivities are reported separately.

The runtime-scoped production study contains 36000 X-channel and 56000 Z-channel paired fault records. All complete recoveries are syndrome-valid, and an independent reconstruction of the checkpoints, joint tables, marginals, paired differences, and final analysis inputs finds zero invalid recoveries. Recorded decoder construction, sampling, and three-protocol replay total 10.734 compute hours.

No primary finite-size fit passes every registered threshold guard. For X noise, full-prefix feedback gives adjacent-size crossings at 5.159%, 5.586%, and 5.684%. Its quadratic diagnostic candidate is $p_c = 5.297\%$ with $\nu = 1.391$, but it is withheld because the deviance 54.77/31 has goodness-of-fit $p = 0.00530 < 0.01$. Accepted cubic and endpoint-window sensitivity fits lie between 5.293% and 5.323%, while a fixed-L^{-1} correction gives 6.154%; these remain model diagnostics because the primary fit fails. The present no-feedback and slice-local X fits are inconclusive in this probability window.

For Z noise, slice-local and full-prefix feedback give diagnostic candidates of 3.934% and 3.923%, respectively. The slice-local data lack the $L = 13, 17$ crossing, the full-prefix data lack the $L = 17, 21$ crossing, and both quadratic fits are nonmonotone with goodness-of-fit below 1.5×10^{-5}. They therefore show a transition-like onset near 3.9% but do not establish an asymptotic threshold. At $L = 21, p = 0.035$, the state-failure rates are 0.7545, 0.2445, and 0.2415 for no feedback, slice-local feedback, and full-prefix feedback. The paired full-prefix-minus-none difference is -0.513 with a conservative 95% interval $[-0.549, -0.475]$, whereas full-prefix minus slice-local is -0.003 with interval $[-0.0146, 0.0086]$. Active Z feedback is therefore effective at fixed size and noise rate, while these data do not resolve a threshold difference between the local and prefix policies.

These withheld results do not supersede the separate targeted estimates above: the earlier low-probability Z final-only study gives 0.517%, and the earlier one-step X slice-local study gives 1.482%. The probability windows in the present three-policy comparison do not refit either estimate. Missing finite-size crossings likewise do not imply a zero threshold. Complete raw counts, Wilson intervals, sensitivity fits, paired statistics, hashes, and reproduction commands are recorded in `toric3d-diagonal-feedback-strength.md` in `docs/superpowers` and its associated results directory.

8 questions

8.1 present 2D toric-code

Can a different unitary extension reduce the propagation of Z errors? Can additional resources overcome the 2D obstruction?

introduce majority-like correction in 2D:

(a) redesigned unitary schedule (changes plaquette order, pivot selection, CNOT directions, CNOT order.)

(b) repeated stabilizer measurement rounds + classical decoding and feedback (measurement feedback)

(c) redundant encoding of virtual information and correct it coherently

8.2 move to 3D toric-code

Use measurement feedback for 3D toric code

prepare one 2D layer → measure the layer syndrome → decode → feedback → prepare the next layer.

compare:

(a) naive 3D sequential circuit

(b) feedback-stabilized 3D sequential circuit

(c) logical failure rate versus noise and system size